

Recovering Motion-Referring Expressions in Moving-Camera Videos via Global Motion Compensation

Anonymous Author(s)

Abstract

Referring multi-object tracking takes a video and a natural-language expression and tracks every object the expression describes. Many expressions refer to how an object moves rather than how it looks, but common hosts read motion from raw box displacement. A moving camera corrupts that signal: a parked car sweeps across the frame and is scored as “moving,” while a car keeping pace with traffic can read as static and be dropped. Motion-language RMOT fuses dynamics with text yet leaves the camera in the signal; ego-motion compensation removes the camera but is used only for association. We connect the two with a decision-level plug-in: per-frame homographies compose into a cumulative transform, residual velocity is read at multiple gaps, a two-tower encoder aligns it with the expression, and the score is added to the host logit without host retraining. On iKUN, aligning motion with language lifts moving-class HOTA by +8.83 (native 20.25 $\rightarrow$ 29.08, $n=5$); most of that lift comes from giving the host any motion descriptor, while ego-compensation is the only engineered component with a significant effect and the only one that serves both directions of the disambiguation: removing it costs -1.77 moving ($p=0.006$) and -2.07 static-class HOTA ($p<10^{-4}$), the parked objects that raw velocity misreads as moving. Across three host-split settings, the recovered signal is monotone-inverse to native motion ability. In pooled HOTA the module matches iKUN’s published score and improves FlexHook’s V2 score outright (+0.281), while FlexHook V1 remains below its published number under a known reproduction gap.

Keywords

referring multi-object tracking, ego-motion compensation, motion-language alignment, plug-in module, Refer-KITTI, video grounding

1 Introduction

Referring multi-object tracking (RMOT) outputs trajectories for the objects described by a free-form language. For example, a language description like *left-vehicles-in-red* mentioned in the Refer-KITTI dataset is used to evaluate the performance of RMOT. Many such descriptions focus on the movement of targeted objects, such as “moving cars” and “a vehicle turning left”, rather than their appearance. However, the motion of objects in a video captured from a moving camera is mixed with the object’s motion and the camera motion. A static car may appear to move in video frames due to camera motion, while a nearby moving car may maintain a nearly fixed distance from the moving camera and appear to be static.

VMMOT [13] and its RMOT lineage [4, 10, 21] align motion with text. But the motion, mixed with object and camera motion, is not specially distinguished. Plain MOT proposes to consider motion compensation. BoT-SORT [1] and UCMCTrack [22] are adopted to estimate camera motion and stabilize predictions, respectively. However, the motion information and language description are not

jointly processed or fused. The motion of the camera or vehicle can be obtained from the global positioning system (GPS) and inertial measurement unit (IMU) annotations provided by KITTI or estimated from image-based methods such as ground-plane projection. Existing RMOT methods, however, do not incorporate this information into language-guided tracking. Instead, our method estimates this motion from image features, removing the need of GPS or IMU sensors. This design also allows the module to be directly integrated into different two-stage RMOT frameworks without modifying their architecture. The only additional requirement is a single scalar calibration for each host model, as described in Section 3.4.

Global Motion Compensation-Link (GMC-Link) separates camera motion from object motion to provide motion information for language-guided tracking. It consists of four stages. First, a homography, which models the geometric transformation between consecutive frames, is estimated to describe camera motion. The estimated homographies are accumulated across multiple frames to compensate for camera motion. Second, residual velocity, which represents object motion after camera motion has been removed, is computed over multiple frames and used as a temporal motion descriptor. Third, the Contrastive Motion Aligner measures how well the motion descriptor matches the language expression. Finally, the resulting motion score is combined with the host model through a calibrated decision-level correction, allowing GMC-Link to improve existing two-stage RMOT frameworks without modifying their architecture.

The accumulated camera transformation is the key component of GMC-Link. By removing camera motion, it allows the motion descriptor to represent the object’s movement rather than the camera’s movement. It also preserves motion information across multiple frames, enabling the model to capture motion patterns described by expressions such as “moving” or “turning”. Based on this design, we hypothesize that camera motion compensation is essential for distinguishing object motion from camera-induced motion. If residual velocity is replaced with raw image velocity, performance should decrease on moving objects while stationary objects are more likely to be misclassified as moving. Section 4 evaluates this hypothesis and shows that residual velocity contributes most of the improvement for moving objects, while camera motion compensation makes the motion descriptor effective for object disambiguation rather than simply providing additional motion cues.

This work connects ego-motion compensation with motion-language modeling for referring multi-object tracking (RMOT). Our main contributions are as follows.

- We propose Global Motion Compensation-Link (GMC-Link), a plug-in module that compensates for camera motion without modifying or retraining the host RMOT model.

- We introduce a motion representation based on residual velocity after camera motion compensation for motion-language alignment.
- We provide experimental evidence showing that camera motion compensation is the key factor for improving motion-aware referring multi-object tracking.
- We demonstrate consistent improvements on iKUN and FlexHook while preserving the original detector, tracker, and appearance-language matching.

GMC-Link consistently improves motion-aware tracking on iKUN and FlexHook while preserving the original detector, tracker, and host architecture.

2 Related Work

2.1 Referring MOT and motion-language fusion

TransRMOT [21] established RMOT on Refer-KITTI, iKUN [4] decoupled grounding from association through a frozen tracker, and FlexHook [10] strengthened the two-stage cross-modal hook. The two-stage referring-by-tracking line continues to grow, with zero-shot MLLM captioning over frozen tracks (ReferGPT [3]) and memory-efficient plug-in referring modules (MEX [19]). Later RMOT methods improve semantic grounding [7, 8, 11, 14]; VM-RMOT [13] even makes motion explicit with position, direction, distance-trend, and speed-trend descriptors. The common limitation is where the dynamics come from: raw image-plane boxes. Thus motion-language RMOT can be language-aware while its motion representation remains entangled with camera ego-motion. TempRMOT [23] already aggregates trajectory history in recurrent memory, and we exclude such hosts on evidence rather than convenience: cascading the module onto TempRMOT regressed pooled HOTA in two independent trials (-6.75 under an early cascade rule, -3.8 to -5.4 across a later coefficient sweep), consistent with an external motion module adding redundant constraints to a host whose recurrent state already encodes trajectory history.

2.2 Camera and ego-motion compensation in tracking

Camera compensation is standard in plain MOT, where it stabilizes predictions before association. BoT-SORT [1] estimates an inter-frame transform from ORB matches [17] with RANSAC [5], while UCMCTrack [22] uses a ground-plane camera-motion model. EMAP [15] goes furthest in this direction, decoupling camera rotation and translation from object motion inside the Kalman prediction itself and improving KITTI HOTA by over 5% across four trackers. All three improve association under ego-motion, but the corrected geometry is consumed immediately by frame-to-frame association. It is not composed into a persistent ego-stable trajectory, accumulated over the temporal span of a motion expression, or exposed as a language-aligned feature. The geometric core we rely on is classical — plane-plus-parallax motion separation [9], homography estimation [6], motion segmentation under a moving camera [2], and ego-motion estimation in visual odometry [18] — and we claim no novelty in the separation itself; the contribution is routing its output to language.

2.3 The unfilled intersection

The gap is specific: motion-language RMOT is language-aware but camera-contaminated; camera-compensated MOT is ego-stable but language-unaware. GMC-Link connects them by producing persistent ego-stable motion, aligning it with the expression, and fusing it after the host score. The method therefore preserves host appearance grounding while supplying the motion evidence that the host never had.

3 Method

GMC-Link sits between a two-stage RMOT host and its scoring step. The host remains responsible for detection, association, and appearance-language grounding. The GMC branch estimates background camera motion and converts host tracks into ego-stable residual object motion. The aligner asks whether that residual motion matches the referring expression, and fusion adds only a calibrated correction after the host has formed its appearance-based score. This modularity enables post-hoc attachment without replacing the detector, retraining host weights, or hiding the source of a correction. Figure 1 gives the attachment point, and Figure 2 shows the four stages.

The separation is also diagnostic: failures can be assigned to host appearance grounding, ego-motion estimation, motion-language alignment, or scalar calibration instead of to an opaque fused representation.

3.1 Ego-motion compensation

The first stage estimates camera motion between adjacent frames. We use ORB as a sparse feature-point sampler because global camera motion should be estimated from repeatable, matchable background evidence rather than from every image location. Dense sampling or dense optical flow supplies many more measurements, but in driving footage many of those measurements come from weak-texture road regions, object boundaries, reflections, residual parallax, and independently moving foreground objects; these ambiguous samples can introduce high-volume noise and bias a single global transform. ORB instead keeps a compact set of discriminative corners with binary descriptors, making descriptor matching and later outlier rejection more reliable and efficient. We detect ORB keypoints [17], match descriptors with a brute-force Hamming matcher, and keep matches passing Lowe’s 0.7 ratio test. Host boxes seed a foreground mask: without masking, correspondences on moving objects can bias the transform away from background camera motion, so keypoints inside tracked objects are suppressed and the surviving matches better represent static scene structure. RANSAC [5] fits $H_{t-1 \rightarrow t}$ and records median inlier reprojection error as background residual. If too few reliable inliers survive, the stage falls back to identity, so a weak-texture frame degrades to “no compensation” rather than to a corrupted camera-motion estimate.

A single homography is exact only for planar or pure-rotation scenes, but driving footage is sufficiently planar-dominated for this approximation to be useful. In early separation trials on our data, ORB with RANSAC outperformed both classical dense flow (Farneback) and a learned flow (RAFT) as the ego-motion source, because explicit feature selection and outlier rejection discard noisy or non-background measurements that a dense field tends to smear

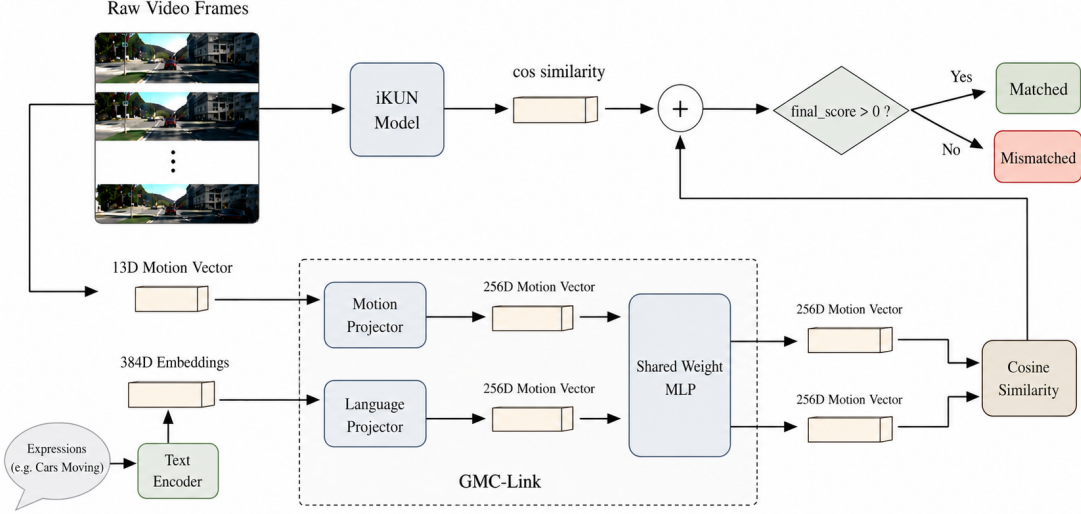


Figure 1: GMC-Link architecture. The host provides tracks and expression scores; GMC-Link estimates camera motion, derives residual motion, aligns it with language, and returns only a calibrated decision-layer correction.

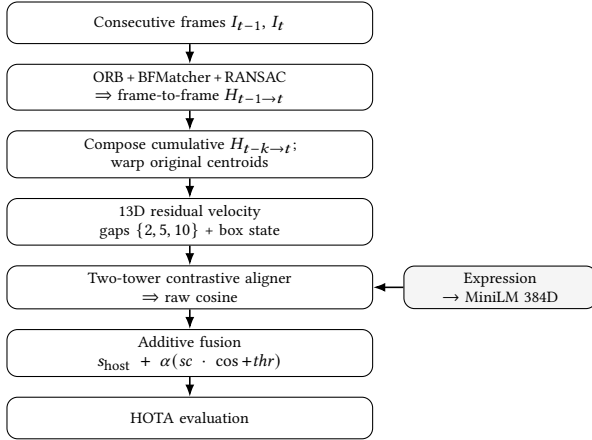


Figure 2: GMC-Link pipeline. ORB homographies compose into a cumulative transform, produce a 13D residual-velocity descriptor, align with the expression, and correct the host logit for HOTA evaluation.

into the estimate. Frame-to-frame compensation is still insufficient for language, because motion expressions span multiple frames. We therefore store the original, never-warped centroids and compose the frame-to-frame transforms into one cumulative homography,

$$H_{t-k \rightarrow t} = H_{t-1 \rightarrow t} H_{t-2 \rightarrow t-1} \cdots H_{t-k \rightarrow t-k+1}, \quad (1)$$

then apply it once to map a past centroid into the current frame. This creates a persistent ego-stable reference while avoiding the error that would accumulate from repeatedly warping already-warped coordinates.

3.2 Multi-scale residual velocity

For temporal gap g , raw velocity is centroid displacement and contains both object motion and camera-induced displacement. Ego velocity estimates the displacement caused by the camera alone by warping the past centroid through Eq. 1. Residual velocity is their difference,

$$v_g^{\text{res}} = v_g^{\text{raw}} - v_g^{\text{ego}}, \quad (2)$$

which approximates the motion a stationary camera would observe. Thus a parked car has near-zero residual motion even as its pixels move significantly under ego-motion. We normalise by image dimensions, $v_{\text{norm}} = (v_{\text{pix}}/d_{\text{img}}) \times 100$.

We compute residuals at gaps $\{2, 5, 10\}$ and concatenate them with box state into a 13D vector ordered as (dx_s, dy_s) , (dx_m, dy_m) , (dx_l, dy_l) , $\Delta w, \Delta h, c_x, c_y, w, h, snr$; the first six entries are residual (dx, dy) pairs at gaps 2, 5, and 10. The scales are complementary rather than redundant. Gap 2 preserves short-term changes and jitter that long windows average away; gap 10 accumulates slow sustained motion that can be buried in one short step; gap 5 bridges the regimes. This matters for language because expressions may refer to instantaneous changes or sustained movement over different temporal spans. The final snr is the mid-scale residual object speed relative to the RANSAC-inlier background noise floor. It does not create class separation itself, but represents homography reliability and reduces frame-level score variance when compensation is weak.

3.3 Motion-language alignment

A two-tower encoder maps the 13D motion vector and expression into a shared space. Linear adapters (motion 13 $\rightarrow$ 256, language 384 $\rightarrow$ 256 from all-MiniLM-L6-v2 [16]) feed a shared MLP

Table 1: Locked fusion coefficients for Eq. 3, hand-set per host-split setting. FlexHook (V1) and (V2) are one architecture calibrated per split. Appearance scale is damped because motion is uninformative on appearance expressions.

Host	Axis	α	sc	thr
iKUN	motion	1.00	0.90	+0.17
	appearance	1.00	0.30	+0.10
FlexHook (V1)	motion	0.65	10.0	+3.0
	appearance	1.00	3.50	+0.9
FlexHook (V2)	motion	0.40	10.0	+1.3
	appearance	1.00	3.50	+1.2

(256 $\rightarrow$ 512 $\rightarrow$ 512 $\rightarrow$ 256), layer norm, and ℓ_2 normalisation. Training uses symmetric InfoNCE [20] at temperature 0.07 with shared-expression false negatives masked. Inference uses raw cosine similarity, with no sigmoid or temporal smoothing, so fusion receives an uncalibrated alignment score. The aligner is intentionally standard: the scientific claim is not a new text encoder or contrastive loss, but that ego-compensated residual motion is the right signal to align. CLIP-text and larger 768D sentence encoders did not consistently improve the shipped raw-cosine pipeline, so MiniLM stays.

3.4 Decision-level additive fusion

The alignment score corrects the host logit at the decision level. We inject it additively per host and per axis,

$$s_{\text{final}} = s_{\text{model}} + \alpha (sc \cdot \cos + thr), \quad (3)$$

where $\cos$ is raw motion-language cosine, sc rescales it to the host logit range, thr shifts the operating point, and α sets weight. We use decision-level fusion because feature-level motion injection into the visual encoder regressed an early pipeline by -21.7% F1; one negative trial does not establish decision-level fusion as optimal, but it motivates keeping motion evidence out of the host’s appearance representation. The final design therefore preserves the host’s appearance representation and injects motion only after the host has formed its score. This keeps appearance grounding intact, prevents motion features from distorting the visual representation, and leaves host evidence and motion evidence as separate auditable terms.

A keyword router selects the motion recipe when an expression contains one of 15 fixed motion substrings (e.g. *moving*, *turning*, *parked*, *braking*); otherwise it selects the appearance recipe. Because motion evidence is largely uninformative for appearance-only expressions, sc_{appear} is seven to eleven times smaller than sc_{motion} . The coefficients absorb host-specific logit-scale differences, so the table should be read as calibration to each host’s score range rather than as a learned fusion head.

Standard-deviation matching and a learned fusion network were tested but underperformed this fixed scalar calibration. This is consistent with the plug-in goal: a new host needs calibration to its score range, but not retraining of its detector, tracker, visual encoder, or language grounding pathway.

4 Experiments

4.1 Setup

We evaluate on Refer-KITTI V1 [21] and V2 [23] using TrackEval HOTA [12]. Moving-class HOTA tests whether the module fixes the target failure; pooled HOTA checks whether that recovery survives the full benchmark. Comparing iKUN [4] with FlexHook [10] tests whether the gain concentrates where native motion reasoning is weakest, and the raw-velocity ablation tests whether ego-compensation matters beyond adding any motion descriptor. FlexHook (V1/V2) is one architecture calibrated per split, not two independent methods. Results are mean $\pm$ std over $n=3$ seeds for cross-setting tables and $n=5$ for ablations with Welch tests; a seed randomizes aligner training (initialization and batch order) only, while homography estimation and host outputs are deterministic. The aligner uses AdamW (10^{-3} , batch 256, 100 epochs); ORB uses 1500 features and RANSAC a 5 px threshold. All iKUN comparisons use the paper-canonical ground-truth template aligned with the public NeuralSORT detections, and each host runs its published public detections unchanged. We report native hosts and a simple uncalibrated raw-cosine GMC control to separate adding the signal from calibrating it to the host logit scale. Code, trained aligner weights for every seed, per-sequence score caches, and the coefficient tables will be released upon publication.

Refer-KITTI has no held-out tuning split, so Table 1’s shipped coefficients are calibrated on the benchmark sequences and should not be claimed as universally transferable hyperparameters. Two observations bound the fitting capacity. First, the scale coefficients were calibrated independently per FlexHook split yet came out identical ($sc=10.0$ motion, 3.5 appearance on both V1 and V2 in Table 1), suggesting the calibration tracks the host’s logit range rather than the particular test sequences; with those shared, roughly four fitted scalars per setting remain against three to four evaluation sequences. Second, we use leave-one-sequence-out calibration to bound overfitting risk directly. iKUN LOSO moving HOTA remains 28.38 ± 1.60 versus 29.47 ± 0.88 in-sample on the same three seeds (native 20.25; the $n=5$ ablation reference in Table 4 is 29.08 ± 0.83), while FlexHook LOSO pooled is 53.28 ± 0.17 (V1) and 42.74 ± 0.07 (V2), with V2 still above the published 42.526. We therefore claim the iKUN moving-class recovery and the FlexHook V2 pooled gain, not the iKUN pooled motion, as calibration-robust.

The test splits are heavily skewed toward appearance. V1 holds 158 expressions over 1010 frames, of which 75.3% are appearance queries, 7.6% are static-motion queries, and only 17.1% (27 expressions) refer to a moving object; V2 holds 862 expressions over 1469 frames with a similar 74.1 / 11.7 / 14.2 split. The motion class our module targets is therefore a roughly one-in-six minority. Pooled HOTA is computed once over the union of all trajectories, so it is dominated by detection and association counts from the appearance majority; it is not a class-weighted mean of per-class HOTA. A recovery concentrated on moving expressions can therefore register as a small pooled movement even when the per-class effect is large. The module’s target failure is motion-class grounding, so moving-class HOTA is the pre-specified effect metric rather than a post-hoc selection; pooled HOTA is the field-standard whole-benchmark summary, reported for every setting with all deltas anchored to the native host.

Table 2: Pooled HOTA on Refer-KITTI test ($n=3$, mean $\pm$ std). Δ is against published pooled HOTA; iKUN matches, while FlexHook V2 is the clean pooled improvement.

Host	Native	+GMC simple	+GMC (ship)	Δ paper
iKUN [4] (V1)	44.564	44.272	44.634 ± 0.066	+0.070
FlexHook (V1) [10]	53.824	53.121	53.526 ± 0.087	$-0.298^\dagger$
FlexHook (V2) [10]	42.526	42.532	42.807 ± 0.038	+0.281

† FlexHook (V1) paper gap is structural (no tested configuration beats 53.824); Δ vs +GMC simple anchor = +0.405.

4.2 Main results

Table 2 first shows that raw GMC evidence is not sufficient by itself: the uncalibrated +GMC simple control is negative on iKUN and FlexHook V1. The signal must be calibrated to the host’s score range. With the shipped calibration, iKUN [4] reaches 44.634 ± 0.066 , matching rather than materially beating the published 44.564, while FlexHook V2 gives the clean pooled improvement, 42.807 ± 0.038 versus 42.526 (+0.281). Pooled parity on iKUN should be read as a guarantee rather than a shortfall: the moving-class recovery is purchased at no cost to the roughly 83% of queries the module does not target, which is the property that makes a plug-in safe to attach. FlexHook V1 improves over +GMC simple ($53.121 \rightarrow 53.526$) but remains below its published 53.824: no tested configuration beats that number — the strongest legacy variant reaches 53.716 ± 0.068 ($n=3$) — so we treat the V1 gap as a reproduction limit of our pipeline and report the module’s V1 gain against the +GMC simple control. The scientific claim is therefore not that GMC-Link beats every pooled leaderboard number; it is that ego-compensated motion recovers motion-class errors, especially where native motion reasoning is weakest. For external context, VMRMOT [13], the strongest motion-language host on this benchmark, reports 53.00/35.21 pooled HOTA on V1/V2 with its own end-to-end detector and tracker. The FlexHook settings with our module reach 53.526/42.807 on the same splits, but we attribute that ordering to host and pipeline rather than to the module and do not claim a head-to-head win; the class-level comparison our claim would require is not computable from published numbers, as VMRMOT does not report per-class moving HOTA. We therefore compare each host against its own native score.

4.3 The motion deficit

The gain tracks native motion deficit. iKUN starts lowest and gains most ($+8.83 \pm 0.83$); FlexHook V1 and V2 start higher and gain little ($+0.91 \pm 0.79$ and $+0.43 \pm 0.17$). Across the three settings, the relation is monotone-inverse: larger native deficit receives larger correction. This supports the intended use case, a host that already grounds appearance but does not reliably distinguish ego-induced displacement from object motion. We treat the pattern as a prediction over three settings — effectively two independent architectures, since the FlexHook pair shares weights-in-kind — not a proportional law; confirming it as a characterization requires further hosts.

Table 3: Motion deficit across host-split settings. Native moving-class HOTA versus the moving-class HOTA gain added by the module. The gain is monotone-inverse to native motion ability across the three settings; the FlexHook rows are one architecture calibrated per split.

Host	Native MOVING	GMC MOVING gain
iKUN [4]	20.25	$+8.83 \pm 0.83$ ($n=5$)
FlexHook (V1) [10]	43.98	$+0.91 \pm 0.79$ ($n=3$)
FlexHook (V2) [10]	48.02	$+0.43 \pm 0.17$ ($n=3$)

Moving-class denominators: 27 of 158 expressions (V1); 14.2% of 862 (V2).

Table 4: Per-class HOTA ablation on iKUN ($n=5$). Ego-compensation is the only significant engineered component, and the only one that moves both classes: removing it costs -1.77 moving (Welch $p=0.006$) and -2.07 static ($p<10^{-4}$).

Variant	MOVING HOTA	STATIC HOTA
Native host (no module)	20.25	—
Full module	29.08 ± 0.83	43.26 ± 0.26
—ego (raw velocity)	27.31 ± 0.66	41.19 ± 0.39
—multiscale (single gap)	28.82 ± 0.84	43.42 ± 0.10
—snr	29.74 ± 0.59	43.35 ± 0.35

Native static-class HOTA under the ablation readout is not archived; deltas among module arms share one convention and are directly comparable.

4.4 Ablations

The ablation directly tests the Introduction hypothesis, and we report the decomposition rather than the headline alone. The full module lifts the moving class $20.25 \rightarrow 29.08$ ($+8.83$); the —ego arm, which feeds raw uncompensated velocity to the same aligner, already reaches 27.31, so having a motion descriptor at all supplies $+7.06$ of the recovery and ego-compensation the remaining $+1.77$ (Welch $t=3.75$, $p=0.006$, 95% CI $[0.67, 2.87]$, Cohen’s $d \approx 2.4$). The moving class understates what compensation buys, because raw velocity errs in both directions: on the static class, the —ego arm drops $43.26 \rightarrow 41.19$, a -2.07 cost ($t=9.81$, $p<10^{-4}$) concentrated exactly on the parked objects whose pixels sweep across the frame. Single-gap motion (-0.26 moving, $p=0.64$) and removing snr ($+0.67$ moving, $p=0.19$) stay within seed spread on both classes, so ego-compensation is the only statistically significant engineered component, and the only one the disambiguation claim rests on; its effect survives Bonferroni correction over the three component tests ($p_{\text{corr}}=0.018$). Two caveats keep the null results honest: at $n=5$ they bound only large effects, so modest multiscale or snr contributions cannot be excluded, and the —snr arm’s higher mean is itself within that spread — snr is retained as the variance-reduction feature of Section 3.2, and dropping it would be a defensible simplification rather than a loss.

Feature-level injection regressed an early host by -21.7% F1, supporting the decision to correct logits after appearance scoring rather than reshaping host representations.

Figure 3 makes the failure mode concrete. The parked car has large raw image displacement because the camera passes it, while



Figure 3: Real recovery under camera ego-motion on “moving cars” (Refer-KITTI seq 0005, frames 85/94/100 of a forward-driving clip; the green car moves, the red car is parked). The dashed line marks the mover’s image column. **Red:** the parked car’s centroid slides left as the camera passes it (center- x 261 $\rightarrow$ 104 px), so the camera-naive host scores it “moving” (false positive), yet its ego-compensated residual is near zero and the module restores it to static. **Green:** the moving car holds its column (center- $x \approx 605$, < 1 px/frame), so the host misses it (false negative), but its nonzero residual — unlike the static background — recovers it as moving. Both errors flip from the same host logit through the additive correction.

the genuinely moving car maintains nearly the same image column. Raw image-plane motion therefore gives the wrong semantic cue in both directions; ego-compensated residual motion restores the distinction. The module runs at 68 FPS (median 14.8 ms/frame excluding host inference), dominated by CPU ORB homography (14.55 ms); the aligner is under 0.2 ms.¹

5 Conclusion and Limitations

A decision-level motion-language module is a targeted correction: it recovers the motion-class referring expressions that camera-naive hosts miss, without touching the appearance majority. The bulk of the recovery comes from aligning a motion descriptor with language at all; ego-compensation is the component that turns the descriptor into a disambiguator, significantly protecting both the moving and the static class. The result suggests that motion-language grounding in moving-camera video should reason over ego-stable trajectories, not raw image-plane displacement. The module keeps the tracker and detector fixed, retrain no host weights, and adds

only host-specific scalar calibration. On iKUN [4] it matches published pooled HOTA while lifting the hidden motion class; on FlexHook [10] V2 it improves pooled HOTA outright.

The limits are bounded. A single ORB/RANSAC homography [1] is approximate under non-planar scenes and strong parallax; our evidence is therefore confined to planar-dominated Refer-KITTI driving footage, with no claim of transfer to handheld, aerial, or strongly non-planar video. A related failure mode is keypoint starvation: a large mover filling the frame leaves little background texture, triggering the identity fallback exactly when compensation is most needed; the snr feature down-weights such frames but does not recover them. Calibration is host-specific and benchmark-level coefficients are not universal hyperparameters. We study two host architectures across three host-split settings, with FlexHook V1/V2 not independent architectural confirmations. Hosts with native temporal memory, such as TempRMOT [23], remain out of scope on tested evidence: cascaded trials regressed pooled HOTA rather than improving it.

The broader implication is that camera compensation can serve not only tracking association, but also language grounding. Richer ego-stable trajectory descriptors are the natural next step for camera-naive RMOT hosts.

References

- [1] Nir Aharon, Roy Orfaig, and Ben-Zion Bobrovsky. 2022. BoT-SORT: Robust Associations Multi-Pedestrian Tracking. arXiv:2206.14651 [cs.CV] doi:10.48550/arXiv.2206.14651
- [2] Pia Bideau and Erik Learned-Miller. 2016. It’s Moving! A Probabilistic Model for Causal Motion Segmentation in Moving Camera Videos. In *Computer Vision – ECCV 2016 (Lecture Notes in Computer Science, Vol. 9912)*. Springer, 433–449. doi:10.1007/978-3-319-46484-8_26
- [3] Tzoulis Chamiti, Leandro Di Bella, Adrian Munteanu, and Nikos Deligiannis. 2025. ReferGPT: Towards Zero-Shot Referring Multi-Object Tracking. arXiv:2504.09195 [cs.CV] CVPR Workshops.
- [4] Yunhao Du, Cheng Lei, Zhicheng Zhao, and Fei Su. 2024. iKUN: Speak to Trackers without Retraining. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 19135–19144. doi:10.1109/CVPR52733.2024.01810
- [5] Martin A. Fischler and Robert C. Bolles. 1981. Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography. *Commun. ACM* 24, 6 (1981), 381–395. doi:10.1145/358669.358692
- [6] Richard Hartley and Andrew Zisserman. 2004. *Multiple View Geometry in Computer Vision* (2nd ed.). Cambridge University Press.
- [7] Wenyan He, Yajun Jian, Yang Lu, and Hanzi Wang. 2024. Visual-Linguistic Representation Learning with Deep Cross-Modality Fusion for Referring Multi-Object Tracking. In *ICASSP 2024 – 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 6310–6314. doi:10.1109/ICASSP48485.2024.10447535
- [8] Wenjun Huang, Yang Ni, Hanning Chen, Yirui He, Ian Bryant, Yezi Liu, and Mohsen Imani. 2024. Tell Me What to Track: Infusing Robust Language Guidance for Enhanced Referring Multi-Object Tracking. arXiv preprint arXiv:2412.12561. arXiv:2412.12561 [cs.CV]
- [9] Michal Irani and P. Anandan. 1998. A Unified Approach to Moving Object Detection in 2D and 3D Scenes. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 20, 6 (1998), 577–589. doi:10.1109/34.683770
- [10] Weize Li, Yunhao Du, Qixiang Yin, Zhicheng Zhao, and Fei Su. 2025. Rethinking Two-Stage Referring-by-Tracking in Referring Multi-Object Tracking: Make it Strong Again. arXiv:2503.07516 [cs.CV] doi:10.48550/arXiv.2503.07516 Accepted to CVPR 2026.
- [11] Shaofeng Liang, Runwei Guan, Wangwang Lian, Daizong Liu, Xiaolou Sun, Dongming Wu, Yutao Yue, Weiping Ding, and Hui Xiong. 2025. Cognitive Disentanglement for Referring Multi-Object Tracking. *Information Fusion* 124 (2025), 103349. doi:10.1016/j.inffus.2025.103349
- [12] Jonathon Luiten, Aljoša Osé, Patrick Dendorfer, Philip Torr, Andreas Geiger, Laura Leal-Taixé, and Bastian Leibe. 2021. HOTA: A Higher Order Metric for Evaluating Multi-Object Tracking. *International Journal of Computer Vision* 129, 2 (2021), 548–578. doi:10.1007/s11263-020-01375-2
- [13] Weiyei Lv, Ning Zhang, Hanyang Sun, Haoran Jiang, Kai Zhao, Jing Xiao, and Dan Zeng. 2025. Vision-Motion-Reference Alignment for Referring Multi-Object

¹An orthogonal appearance re-ranker reaches 45.612 pooled HOTA on iKUN ($n=3$); we do not claim it here.

- Tracking via Multi-Modal Large Language Models. arXiv:2511.17681 [cs.CV] doi:10.48550/arXiv.2511.17681
- [14] Zeliang Ma, Song Yang, Zhe Cui, Zhicheng Zhao, Fei Su, Delong Liu, and Jingyu Wang. 2024. MLS-Track: Multilevel Semantic Interaction in RMOT. arXiv preprint arXiv:2404.12031. arXiv:2404.12031 [cs.CV]
 - [15] Navid Mahdian, Mohammad Jani, Amir M. Soufi Enayati, and Homayoun Najjarian. 2024. Ego-Motion Aware Target Prediction Module for Robust Multi-Object Tracking. arXiv:2404.03110 [cs.CV]
 - [16] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Association for Computational Linguistics, Hong Kong, China, 3982–3992. doi:10.18653/v1/D19-1410
 - [17] Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. 2011. ORB: An Efficient Alternative to SIFT or SURF. In *2011 International Conference on Computer Vision (ICCV)*. IEEE, 2564–2571. doi:10.1109/ICCV.2011.6126544
 - [18] Davide Scaramuzza and Friedrich Fraundorfer. 2011. Visual Odometry [Tutorial]. *IEEE Robotics & Automation Magazine* 18, 4 (2011), 80–92. doi:10.1109/MRA.2011.943233
 - [19] Huu-Thien Tran, Phuoc-Sang Pham, Thai-Son Tran, and Khoa Luu. 2025. MEX: Memory-efficient Approach to Referring Multi-Object Tracking. arXiv:2502.13875 [cs.CV]
 - [20] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. 2018. Representation Learning with Contrastive Predictive Coding. arXiv:1807.03748 [cs.LG]
 - [21] Dongming Wu, Wencheng Han, Tiancai Wang, Xingping Dong, Xiangyu Zhang, and Jianbing Shen. 2023. Referring Multi-Object Tracking. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 14633–14642. doi:10.1109/CVPR52729.2023.01406
 - [22] Kefu Yi, Kai Luo, Xiaolei Luo, Jiangui Huang, Hao Wu, Rongdong Hu, and Wei Hao. 2024. UCMCTrack: Multi-Object Tracking with Uniform Camera Motion Compensation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. AAAI Press, 6702–6710. doi:10.1609/aaai.v38i7.28493
 - [23] Yani Zhang, Dongming Wu, Wencheng Han, and Xingping Dong. 2024. Bootstrapping Referring Multi-Object Tracking. arXiv:2406.05039 [cs.CV] doi:10.48550/arXiv.2406.05039 Introduces TempRMOT and Refer-KITTI-V2.